Part 1: Algorithm Details

Algorithm Name

Extra-Gradient Method (EG) for Convex-Concave Saddle Point Problems

We consider the saddle point (min-max) problem

$$\min_{x \in \mathbb{R}^n} \max_{y \in \mathbb{R}^m} L(x, y),$$

where L is convex in x and concave in y. (We treat the $\max_x \min_y$ formulation symmetrically; by sign convention this is the same as the standard $\min_x \max_y$ analysis.)

Mathematical Formulation

Let $z = (x, y) \in \mathbb{R}^{n+m}$ be the joint variable. Define the *monotone operator* (saddle gradient field)

$$F(z) = \begin{pmatrix} \nabla_x L(x, y) \\ -\nabla_y L(x, y) \end{pmatrix}.$$

A saddle point $z^\star = (x^\star, y^\star)$ satisfies the variational inequality

$$\langle F(z^\star), z - z^\star \rangle \geq 0 \quad \forall z.$$

Update rules (operator form). With step size $\eta > 0$, the Extra-Gradient method performs:

$$z_{k+1/2} = z_k - \eta\, F(z_k) \qquad \text{(extrapolation / prediction step)} \tag{1}$$

$$z_{k+1} = z_k - \eta\, F(z_{k+1/2}) \qquad \text{(correction step using the midpoint gradient).} \tag{2}$$

Update rules (componentwise).

$$x_{k+1/2} = x_k - \eta\, \nabla_x L(x_k, y_k), \qquad y_{k+1/2} = y_k + \eta\, \nabla_y L(x_k, y_k), \tag{3}$$

$$x_{k+1} = x_k - \eta\, \nabla_x L(x_{k+1/2}, y_{k+1/2}), \qquad y_{k+1} = y_k + \eta\, \nabla_y L(x_{k+1/2}, y_{k+1/2}). \tag{4}$$

Hyperparameters.

- Step size $\eta > 0$, required to satisfy $\eta \leq \frac{1}{L_{\mathrm{Lip}}}$ where L_{Lip} is the Lipschitz constant of F.
- Number of iterations K.
- The averaged iterate $\bar{z}_K = \frac{1}{K} \sum_{k=0}^{K-1} z_{k+1/2}$ is returned (ergodic averaging).

Pseudocode

```
Input: initial point z_0 = (x_0, y_0), step size eta, iterations K
Initialize: S = 0   (running sum for averaging)
for k = 0, 1, ..., K-1 do
    # Extrapolation step
    g_k       = F(z_k)                 # F(z) = (grad_x L, -grad_y L)
    z_{k+1/2} = z_k - eta * g_k
    # Correction step
    g_mid     = F(z_{k+1/2})
    z_{k+1}   = z_k - eta * g_mid
    # Accumulate for ergodic average
```

```
    S = S + z_{k+1/2}
    # Termination check
    if ||g_mid|| <= tol then break
end for
Output: averaged iterate  z_bar = S / K   (or last z_{k+1})
```

Dry Run Example

To illustrate the mechanics on a scalar example we use the convex function $f(w) = (w-3)^2$ (a pure minimization, i.e. the y-block is absent and $F(w) = f'(w) = 2(w-3)$). Parameters: $w_0 = 0$, $\eta = 0.1$, 3 iterations.

Iteration $k = 0$.

$$\begin{aligned} g_0 &= f'(w_0) = 2(0-3) = -6, \\ w_{1/2} &= w_0 - \eta g_0 = 0 - 0.1(-6) = 0.6, \\ g_{\text{mid}} &= f'(w_{1/2}) = 2(0.6-3) = -4.8, \\ w_1 &= w_0 - \eta g_{\text{mid}} = 0 - 0.1(-4.8) = 0.48, \\ f(w_1) &= (0.48-3)^2 = 6.3504. \end{aligned}$$

Iteration $k = 1$.

$$\begin{aligned} g_1 &= f'(w_1) = 2(0.48-3) = -5.04, \\ w_{3/2} &= w_1 - \eta g_1 = 0.48 - 0.1(-5.04) = 0.984, \\ g_{\text{mid}} &= f'(w_{3/2}) = 2(0.984-3) = -4.032, \\ w_2 &= w_1 - \eta g_{\text{mid}} = 0.48 - 0.1(-4.032) = 0.8832, \\ f(w_2) &= (0.8832-3)^2 = 4.4808. \end{aligned}$$

Iteration $k = 2$.

$$\begin{aligned} g_2 &= f'(w_2) = 2(0.8832-3) = -4.2336, \\ w_{5/2} &= w_2 - \eta g_2 = 0.8832 - 0.1(-4.2336) = 1.30656, \\ g_{\text{mid}} &= f'(w_{5/2}) = 2(1.30656-3) = -3.38688, \\ w_3 &= w_2 - \eta g_{\text{mid}} = 0.8832 - 0.1(-3.38688) = 1.221888, \\ f(w_3) &= (1.221888-3)^2 \approx 3.16111. \end{aligned}$$

The objective decreases monotonically toward the optimum $w^\star = 3$, $f(w^\star) = 0$.

Part 2: Convergence Theory

Assumptions

Assumption 1 (Convexity-Concavity). *$L(x, y)$ is convex in x for every fixed y, and concave in y for every fixed x. Equivalently the operator F is* monotone*:*

$$\langle F(z) - F(z'),\, z - z' \rangle \geq 0 \qquad \forall z, z'.$$

Assumption 2 (Lipschitz smoothness). *F is L_{Lip}-Lipschitz continuous:*

$$\|F(z) - F(z')\| \leq L_{\text{Lip}}\, \|z - z'\| \qquad \forall z, z'.$$

Assumption 3 (Existence of a saddle point)**.** *There exists* $z^\star = (x^\star, y^\star)$ *such that* $L(x^\star, y) \le L(x^\star, y^\star) \le L(x, y^\star)$ *for all* x, y*. Equivalently* $\langle F(z^\star), z - z^\star\rangle \ge 0$ *for all* z*.*

Assumption 4 (Step size)**.** *The step size satisfies* $0 < \eta \le \dfrac{1}{L_{\mathrm{Lip}}}$*.*

We measure convergence through the *primal-dual gap*

$$\mathrm{Gap}(\bar{x}, \bar{y}) \;=\; \max_y L(\bar{x}, y) - \min_x L(x, \bar{y}).$$

Main Convergence Theorem

Theorem 1 (Ergodic $O(1/K)$ rate)**.** *Under Assumptions 1–4, let the iterates be generated by the Extra-Gradient method and define the ergodic average* $\bar{z}_K = \frac{1}{K}\sum_{k=0}^{K-1} z_{k+1/2}$*,* $\bar{z}_K = (\bar{x}_K, \bar{y}_K)$*. Then for any compact feasible set of radius* D *around* $z^\star$*,*

$$\mathrm{Gap}(\bar{x}_K, \bar{y}_K) \;\le\; \frac{\|z_0 - z^\star\|^2}{2\eta K} \;=\; O\!\left(\frac{1}{K}\right).$$

Moreover the iterates remain bounded: $\|z_{k+1} - z^\star\| \le \|z_k - z^\star\|$*.*

Theoretical Properties

- **Stability.** Each step is non-expansive toward $z^\star$ (Lemma **??**); unlike plain Gradient Descent-Ascent (GDA), which can diverge on bilinear games, EG is stable.
- **Rate.** The ergodic gap converges as $O(1/K)$, optimal for monotone Lipschitz problems with first-order oracles.
- **Hyperparameter sensitivity.** Convergence requires $\eta \le 1/L_{\mathrm{Lip}}$. Larger η may violate the descent inequality [Step 7] and break monotonicity.
- **Comparison to baselines.** GDA achieves only $O(\eta)$ neighborhood convergence (or diverges); EG's extrapolation cancels the rotational component of the dynamics, recovering true convergence.

Discussion

The extra (midpoint) gradient evaluation gives EG an implicit look-ahead, effectively approximating the proximal point method to second order. This yields global convergence on the entire monotone class at the cost of two gradient evaluations per iteration.

Part 3: Complete Convergence Proof

Preliminaries (Definitions and Lemmas)

We repeatedly use the **three-point identity**: for any vectors a, b, c,

$$2\langle a - b,\, a - c\rangle = \|a - b\|^2 + \|a - c\|^2 - \|b - c\|^2. \tag{5}$$

Lemma 1 (Monotonicity from convexity-concavity)**.** *If* L *is convex-concave then* F *is monotone, i.e.* $\langle F(z) - F(z'), z - z'\rangle \ge 0$*.*

Proof. Write $z = (x, y)$, $z' = (x', y')$. Convexity in x gives $\langle \nabla_x L(x, y) - \nabla_x L(x', y),\, x - x'\rangle \ge 0$ and similarly along y using concavity (note the sign flip in F's second block makes $-\nabla_y L$ monotone). Summing the blocks yields the claim. □

Lemma 2 (Optimality of $z^\star$). *$\langle F(z^\star),\, z - z^\star\rangle \ge 0$ for all z, and by monotonicity $\langle F(z),\, z - z^\star\rangle \ge 0$ for all z.*

Main Proof (Step-by-Step Derivation)

Fix an arbitrary comparison point z (later set to $z^\star$ or used in the gap argument). Denote $w_k = z_{k+1/2}$ for brevity. Recall:

$$w_k = z_k - \eta F(z_k), \qquad z_{k+1} = z_k - \eta F(w_k).$$

[Step 1] Three-point identity on the correction step. Apply (5) with $a = z_{k+1}$, $b = z_k$, $c = z$. But it is cleaner to expand the squared distance directly. Using $z_{k+1} = z_k - \eta F(w_k)$,

$$\begin{aligned}\|z_{k+1} - z\|^2 &= \|z_k - \eta F(w_k) - z\|^2 \\ &= \|z_k - z\|^2 - 2\eta\langle F(w_k),\, z_k - z\rangle + \eta^2\|F(w_k)\|^2. \end{aligned} \tag{6}$$

[Step 2] Decompose the inner product through w_k. Write $z_k - z = (z_k - w_k) + (w_k - z)$, so

$$\langle F(w_k),\, z_k - z\rangle = \langle F(w_k),\, w_k - z\rangle + \langle F(w_k),\, z_k - w_k\rangle. \tag{7}$$

[Step 3] Substitute (7) into (6).

$$\begin{aligned}\|z_{k+1} - z\|^2 = \|z_k - z\|^2 &- 2\eta\langle F(w_k),\, w_k - z\rangle \\ &- 2\eta\langle F(w_k),\, z_k - w_k\rangle + \eta^2\|F(w_k)\|^2. \end{aligned} \tag{8}$$

[Step 4] Handle the prediction step. From $w_k = z_k - \eta F(z_k)$ we have $z_k - w_k = \eta F(z_k)$. Hence

$$-2\eta\langle F(w_k),\, z_k - w_k\rangle = -2\eta\langle F(w_k),\, \eta F(z_k)\rangle = -2\eta^2\langle F(w_k),\, F(z_k)\rangle. \tag{9}$$

[Step 5] Combine the two η^2 terms. Insert (9) into (8):

$$\begin{aligned}\|z_{k+1} - z\|^2 = \|z_k - z\|^2 &- 2\eta\langle F(w_k),\, w_k - z\rangle \\ &+ \eta^2\|F(w_k)\|^2 - 2\eta^2\langle F(w_k), F(z_k)\rangle. \end{aligned} \tag{10}$$

Complete the square on the last two η^2 terms by adding and subtracting $\eta^2\|F(z_k)\|^2$:

$$\eta^2\|F(w_k)\|^2 - 2\eta^2\langle F(w_k), F(z_k)\rangle = \eta^2\|F(w_k) - F(z_k)\|^2 - \eta^2\|F(z_k)\|^2. \tag{11}$$

[Step 6] Re-express $\eta^2\|F(z_k)\|^2$ via the prediction step. Since $w_k - z_k = -\eta F(z_k)$, we have $\|w_k - z_k\|^2 = \eta^2\|F(z_k)\|^2$. Substituting (11) and this identity into (10):

$$\begin{aligned}\|z_{k+1} - z\|^2 = \|z_k - z\|^2 &- 2\eta\langle F(w_k),\, w_k - z\rangle \\ &+ \eta^2\|F(w_k) - F(z_k)\|^2 - \|w_k - z_k\|^2. \end{aligned} \tag{12}$$

[Step 7] Bound the error term using Lipschitzness. By Assumption 2, $\|F(w_k) - F(z_k)\| \le L_{\mathrm{Lip}}\|w_k - z_k\|$, so

$$\eta^2\|F(w_k) - F(z_k)\|^2 \le \eta^2 L_{\mathrm{Lip}}^2\|w_k - z_k\|^2.$$

Therefore the last two terms of (12) satisfy

$$\eta^2\|F(w_k) - F(z_k)\|^2 - \|w_k - z_k\|^2 \le -\big(1 - \eta^2 L_{\mathrm{Lip}}^2\big)\,\|w_k - z_k\|^2 \;\le\; 0, \tag{13}$$

where the final inequality uses $\eta \le 1/L_{\mathrm{Lip}}$ (Assumption 4), giving $1 - \eta^2 L_{\mathrm{Lip}}^2 \ge 0$.

[Step 8] Key descent inequality. Combining (12) and (13),

$$\|z_{k+1} - z\|^2 \le \|z_k - z\|^2 - 2\eta\langle F(w_k),\, w_k - z\rangle - \big(1 - \eta^2 L_{\mathrm{Lip}}^2\big)\|w_k - z_k\|^2. \tag{14}$$

[Step 9] Non-expansiveness (set $z = z^\star$). By Lemma 2 with monotonicity, $\langle F(w_k),\, w_k - z^\star\rangle \ge \langle F(z^\star), w_k - z^\star\rangle \ge 0$. Hence from (14),

$$\|z_{k+1} - z^\star\|^2 \le \|z_k - z^\star\|^2, \tag{15}$$

proving the iterates are bounded (stability claim of Theorem 1).

[Step 10] Convert the operator inner product to a gap term. By monotonicity, for arbitrary z,

$$\langle F(w_k),\, w_k - z\rangle \ge \langle F(z),\, w_k - z\rangle.$$

For the saddle problem, the right-hand side relates to the function gap. Specifically, with $z = (x, y)$ and $w_k = (x_{k+1/2}, y_{k+1/2})$, convexity-concavity gives

$$\langle F(w_k),\, w_k - z\rangle \;\ge\; L(x_{k+1/2}, y) - L(x, y_{k+1/2}). \tag{16}$$

This is the standard variational-inequality lower bound: convexity in x gives $L(x_{k+1/2}, y) - L(x, y) \le \langle \nabla_x L(w_k), x_{k+1/2} - x\rangle$ and concavity in y gives the matching y-term; summing yields (16).

[Step 11] Rearrange (14) **and drop the negative term.** Using (16) and dropping the non-positive last term of (14):

$$2\eta\big(L(x_{k+1/2}, y) - L(x, y_{k+1/2})\big) \le \|z_k - z\|^2 - \|z_{k+1} - z\|^2. \tag{17}$$

[Step 12] Telescoping over $k = 0, \dots, K-1$. Sum (17); the right side telescopes:

$$2\eta \sum_{k=0}^{K-1} \big(L(x_{k+1/2}, y) - L(x, y_{k+1/2})\big) \le \|z_0 - z\|^2 - \|z_K - z\|^2 \le \|z_0 - z\|^2. \tag{18}$$

[Step 13] Jensen's inequality (convexity-concavity of averages). Let $\bar{x}_K = \frac{1}{K}\sum_k x_{k+1/2}$, $\bar{y}_K = \frac{1}{K}\sum_k y_{k+1/2}$. By convexity in x and concavity in y,

$$L(\bar{x}_K, y) \le \frac{1}{K}\sum_k L(x_{k+1/2}, y), \qquad L(x, \bar{y}_K) \ge \frac{1}{K}\sum_k L(x, y_{k+1/2}).$$

Therefore

$$L(\bar{x}_K, y) - L(x, \bar{y}_K) \le \frac{1}{K}\sum_{k=0}^{K-1} \big(L(x_{k+1/2}, y) - L(x, y_{k+1/2})\big). \tag{19}$$

[Step 14] Combine (18) **and** (19)**.**

$$L(\bar{x}_K, y) - L(x, \bar{y}_K) \le \frac{\|z_0 - z\|^2}{2\eta K}. \tag{20}$$

[Step 15] Take the supremum to form the gap. Maximizing over y and minimizing over x on a ball of radius D about $z^\star$ (so $\|z_0 - z\|^2 \le \|z_0 - z^\star\|^2 + \text{const}$; in the unconstrained case set $z = z^\star$ for each block), we obtain

$$\mathrm{Gap}(\bar{x}_K, \bar{y}_K) = \max_y L(\bar{x}_K, y) - \min_x L(x, \bar{y}_K) \le \frac{\|z_0 - z^\star\|^2}{2\eta K}. \tag{21}$$

This proves Theorem 1. □

Rate Derivation (With Constants)

Choosing the largest admissible step $\eta = 1/L_{\text{Lip}}$ in (21):

$$\text{Gap}(\bar{x}_K, \bar{y}_K) \leq \frac{L_{\text{Lip}} \, \|z_0 - z^\star\|^2}{2K} = O\!\left(\frac{L_{\text{Lip}} D_0^2}{K}\right), \tag{22}$$

where $D_0 = \|z_0 - z^\star\|$. Thus to reach gap $\leq \varepsilon$ one needs

$$K \geq \frac{L_{\text{Lip}} D_0^2}{2\varepsilon}$$

iterations, i.e. $O(1/\varepsilon)$ complexity.

Special Cases

Corollary 1 (Pure minimization)**.** *If $L(x, y) = f(x)$ does not depend on y and f is convex L_{Lip}-smooth, then $F = \nabla f$ and* (21) *reduces to*

$$f(\bar{x}_K) - f(x^\star) \leq \frac{\|x_0 - x^\star\|^2}{2\eta K} = O(1/K),$$

recovering the standard $O(1/K)$ rate for smooth convex minimization.

Corollary 2 (Bilinear games)**.** *For $L(x, y) = x^\top A y$ we have $F(z) = (Ay, -A^\top x)$, which is monotone ($\langle F(z) - F(z'), z - z' \rangle = 0$) and Lipschitz with $L_{\text{Lip}} = \|A\|$. EG converges at $O(1/K)$ on the gap, whereas plain GDA diverges (its iterates spiral outward). This is the canonical example showing the necessity of the extrapolation step.*

Corollary 3 (Strongly monotone case)**.** *If in addition F is μ-strongly monotone ($\langle F(z) - F(z'), z - z' \rangle \geq \mu \|z - z'\|^2$), then* (14) *strengthens to a contraction*

$$\|z_{k+1} - z^\star\|^2 \leq (1 - c) \, \|z_k - z^\star\|^2$$

for some $c = c(\mu, \eta) \in (0, 1)$, yielding linear (geometric) convergence $\|z_K - z^\star\|^2 \leq (1 - c)^K \|z_0 - z^\star\|^2$.